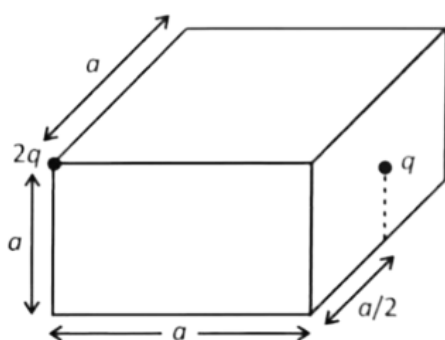


JEE-Main-23-01-2026 (Memory Based)
[MORNING SHIFT]
Physics

Question: A point charge $+2q$ is placed at a vertex of a cube and another point charge $+q$ is placed at the centre of one face of the same cube (both charges lie on the surface). The net electric flux through the cube is:

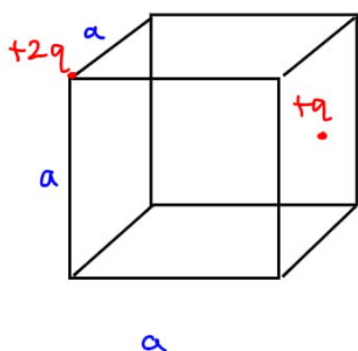


Options:

- (a) $3q/\epsilon_0$
- (b) q/ϵ_0
- (c) $3q/4\epsilon_0$
- (d) $5q/\epsilon_0$

Answer: (c)

Solution:



For a charge on the surface, only a fraction of its total flux contributes.

- Charge $+2q$ at a **vertex**: cube encloses $1/8$ of space around it

$$\Phi_1 = \frac{2q}{8\epsilon_0} = \frac{q}{4\epsilon_0}$$

- Charge $+q$ at **centre of a face**: cube encloses $1/2$ of space around it

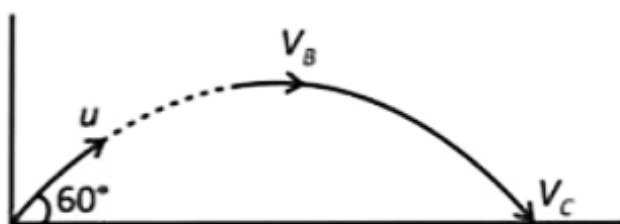
$$\Phi_2 = \frac{q}{2\epsilon_0}$$

Total flux:

$$\Phi = \Phi_1 + \Phi_2 = \frac{q}{4\epsilon_0} + \frac{q}{2\epsilon_0} = \frac{q}{4\epsilon_0} + \frac{2q}{4\epsilon_0} = \frac{3q}{4\epsilon_0}$$

$$\Phi = \frac{3q}{4\epsilon_0}$$

Question: If a projectile is being projected with speed v at angle 60° with horizontal. Find the ratio of speed at highest point (V_B) to the speed at final point (V_C).



Options:

- (a) 3 : 4
- (b) 1 : 3
- (c) 1 : 2
- (d) 1 : 12

Answer: (c)

Solution:

Speed at highest point is $v \cos 60^\circ = \frac{v}{2}$

Speed at C = v

$$\Rightarrow \frac{\frac{v}{2}}{v}$$

$$\therefore \frac{v}{2} \times \frac{1}{v}$$

$$\therefore \frac{1}{2}$$

Question: For the given set of measurement find relative error.

20.00, 19.75, 18.25, 17.01.

Options:

- (a) 0.12
- (b) 0.06
- (c) 0.09
- (d) 0.17

Answer: (b)

Solution:

$$\text{Mean Error} = \frac{20.00 + 19.75 + 18.25 + 17.01}{4}$$

$$= 18.75$$

$$\text{Absolute Error, } \Delta T_1 = 18.75 - 20.00 = -1.25$$

$$\Delta T_2 = 18.75 - 19.75 = -1.00$$

$$\Delta T_3 = 18.75 - 18.25 = +0.50$$

$$\Delta T_4 = 18.75 - 17.01 = +1.74$$

$$\text{Mean Absolute Error, } \Delta T_m = \frac{\sum |\Delta T|}{4}$$

$$= 1.12$$

$$\text{Relative Error} = \frac{1.12}{18.75} = 0.06$$

Question: Find out the correct energy for the ground state or energy transition. (symbols have usual meaning & $n \rightarrow m$ gives the transition).

Options:

- (a) H(-6.8 eV)
- (b) Li^{2+} (-13.6 eV)
- (c) $\text{He}_{2 \rightarrow 1}^+$ (40.8 eV)
- (d) $\text{Be}_{2 \rightarrow 1}^{3+}$ (+13.6 eV)

Answer: (c)

Solution:

Solⁿ Energy in ground state of H: -13.6 eV

Energy for Li^{2+} : $E_1 \times \frac{Z^2}{n^2}$

$$= -13.6 \times \frac{(3)^2}{(1)^2}$$

$$= -122.4 \text{ eV}$$

Energy for He^{2+} : $-13.6 \times \frac{(2)^2}{(1)^2}$

$$= -54.4 \text{ eV}$$

Energy for Be^{3+} transition from 2 to 1

$$E_2 = -13.6 \times \frac{(4)^2}{(2)^2} = -54.4 \text{ eV}$$

$$E_1 = -13.6 \times \frac{(4)^2}{(1)^2} = -217.6 \text{ eV}$$

Question: Two thin prisms are placed in contact such that prism P_2 is inverted relative to prism P_1 . Prism P_1 is made of crown glass with refractive index $\mu_c = 1.50$ (for the given monochromatic light) and has apex angle $A_c = 5^\circ$. Prism P_2 is made of flint glass with refractive index $\mu_f = 1.625$ for the same light. For no net deviation of the emergent ray, the required apex angle A_f of prism P_2 is:

Options:

- (a) 4°
- (b) 5°
- (c) 6°
- (d) 3°

Answer: (a)

Solution:

Concise Solution

For a thin prism, deviation $\delta \approx (\mu - 1)A$.

Since P_2 is inverted relative to P_1 , their deviations oppose each other. For zero net deviation:

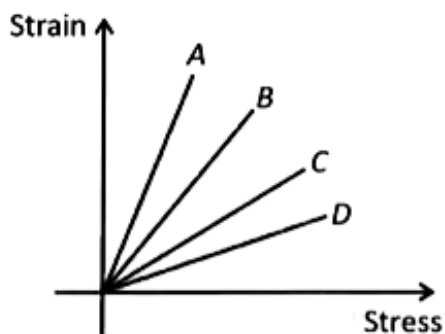
$$(\mu_c - 1)A_c = (\mu_f - 1)A_f$$

$$(1.50 - 1) \cdot 5 = (1.625 - 1) \cdot A_f$$

$$0.50 \times 5 = 0.625 A_f \Rightarrow A_f = \frac{2.5}{0.625} = 4^\circ$$

Question: In the figure, strain is plotted on the y-axis and stress on the x-axis for four materials A, B, C, D. All four lines pass through the origin and represent the linear (elastic) region.

Which material has the greatest Young's modulus?



Options:

- (a) A
- (b) B
- (c) C
- (d) D

Answer: (d)

Solution:

Concise Solution

In elastic limit:

$$\text{strain} = \frac{\text{stress}}{Y} \Rightarrow \frac{\text{strain}}{\text{stress}} = \frac{1}{Y}$$

So, **slope of (strain vs stress) graph** = $\frac{1}{Y}$.

Greater $Y \Rightarrow$ **smaller slope** (less strain for same stress).

Line **D** has the **smallest slope**, hence **maximum Young's modulus**.

Question: A square frame is formed by joining four identical solid cylinders as its sides. Each cylinder has length L , mass M , and radius R . Find the moment of inertia of the frame about the axis lying in the plane of the frame and joining the centers of two opposite cylinders.

Options:

- (a) $I = \frac{2}{3}ML^2 + \frac{3}{2}MR^2$
- (b) $I = \frac{1}{3}ML^2 + \frac{3}{2}MR^2$
- (c) $I = \frac{2}{3}ML^2 + \frac{1}{2}MR^2$
- (d) $I = \frac{1}{3}ML^2 + \frac{1}{2}MR^2$

Answer: (a)

Solution:

1) Two rods perpendicular to the axis (top & bottom):

Axis passes through their centers and is $\perp$ to their lengths:

$$I_1 = 2 \cdot \frac{1}{12} M(L^2 + 3R^2) = \frac{1}{6} ML^2 + \frac{1}{2} MR^2$$

2) Two rods parallel to the axis (left & right):

Use parallel-axis theorem with distance $d = L/2$:

$$I_{\text{each}} = \frac{1}{2} MR^2 + M \left(\frac{L}{2} \right)^2 = \frac{1}{2} MR^2 + \frac{1}{4} ML^2$$

$$I_2 = 2 \left(\frac{1}{2} MR^2 + \frac{1}{4} ML^2 \right) = MR^2 + \frac{1}{2} ML^2$$

Total:

$$I = I_1 + I_2 = \left(\frac{1}{6} + \frac{1}{2} \right) ML^2 + \left(\frac{1}{2} + 1 \right) MR^2 = \frac{2}{3} ML^2 + \frac{3}{2} MR^2$$

So, **option (A)** is correct.

Question: Find the critical angle when light travels from medium A to medium B. Speed of light in A & B is 2.4×10^8 & 2.7×10^8 respectively.

Options:

(a) $\sin^{-1} \left(\frac{2.7}{2.4} \right)$

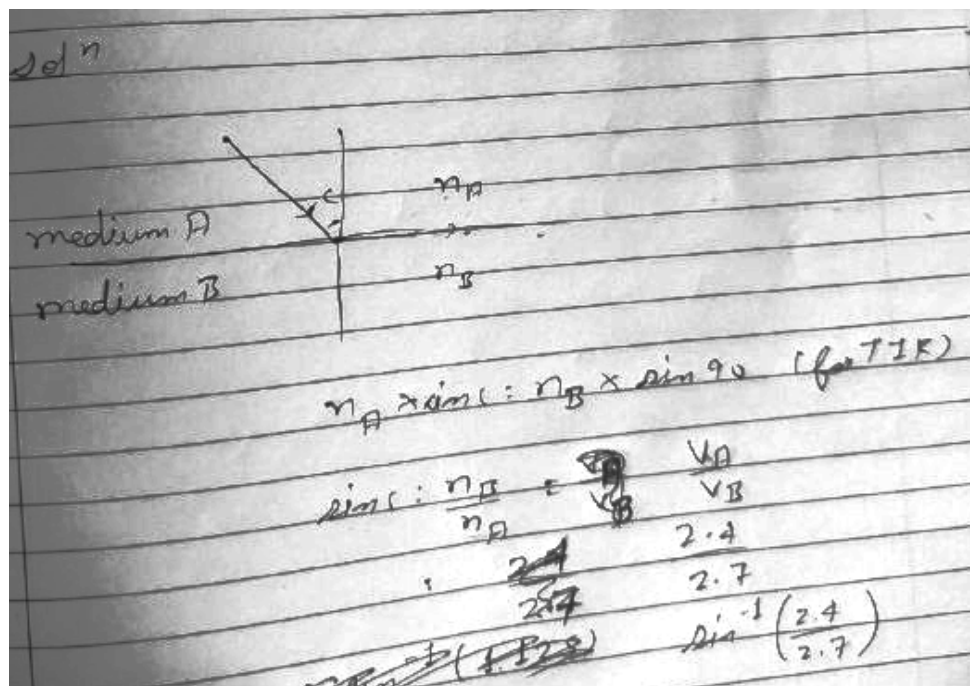
(b) $\sin^{-1} \left(\frac{2.4}{2.7} \right)$

(c) $\tan^{-1} \left(\frac{2.7}{2.4} \right)$

(d) $\tan^{-1} \left(\frac{2.4}{2.7} \right)$

Answer: (b)

Solution:



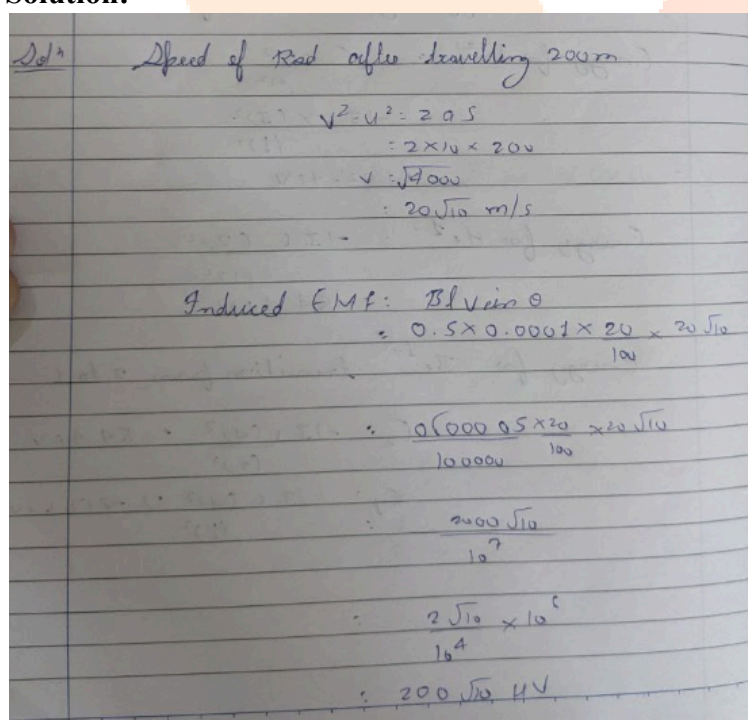
Question: A uniform rod of 20 cm length placed horizontally in a uniform magnetic field 0.5 Gauss perpendicular to its length, the rod has then dropped find the induced emf when it has travelled 200 m.

Options:

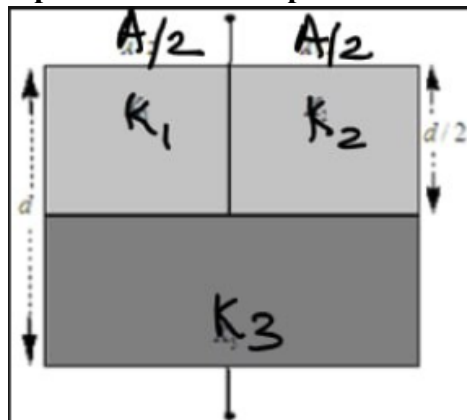
- (a) $200\sqrt{10} \mu\text{V}$
- (b) $20\sqrt{10} \text{ V}$
- (c) $2\sqrt{10} \text{ mV}$
- (d) $2\sqrt{10} \text{ V}$

Answer: (a)

Solution:



Question: A parallel-plate capacitor has plate area A and separation d . The region between the plates is filled with three dielectric slabs arranged exactly as shown in the given diagram. If the dielectric constants are $K_1 = 3$, $K_2 = 5$, $K_3 = 4$, then the equivalent capacitance of the capacitor is:



Options:

- (a) $\frac{8}{3} \frac{\epsilon_0 A}{d}$
 (b) $\frac{16}{5} \frac{\epsilon_0 A}{d}$
 (c) $4 \frac{\epsilon_0 A}{d}$
 (d) $\frac{12}{5} \frac{\epsilon_0 A}{d}$

Answer: (c)

Solution:

Concise Solution

From the diagram, the **top half** consists of two parts (each area $A/2$, thickness $d/2$) in **parallel**:

$$C_1 = \epsilon_0 K_1 \frac{A/2}{d/2} = K_1 \frac{\epsilon_0 A}{d} = 3 \frac{\epsilon_0 A}{d}$$

$$C_2 = \epsilon_0 K_2 \frac{A/2}{d/2} = K_2 \frac{\epsilon_0 A}{d} = 5 \frac{\epsilon_0 A}{d}$$

$$C_{\text{top}} = C_1 + C_2 = 8 \frac{\epsilon_0 A}{d}$$

The **bottom half** (area A , thickness $d/2$):

$$C_3 = \epsilon_0 K_3 \frac{A}{d/2} = 2K_3 \frac{\epsilon_0 A}{d} = 8 \frac{\epsilon_0 A}{d}$$

Now C_{top} and C_3 are in **series**:

$$C_{\text{eq}} = \frac{C_{\text{top}} C_3}{C_{\text{top}} + C_3} = \frac{8 \times 8 \frac{\epsilon_0 A}{d}}{8 + 8} = 4 \frac{\epsilon_0 A}{d}$$

$$\boxed{C_{\text{eq}} = 4 \frac{\epsilon_0 A}{d}}$$

Question: In an LCR series circuit voltage source was given by $V = 5 \sin 100 t$. $I_{\max}$ was 50 mA. Find the value of capacitance is? given $R = 100\Omega$, $L = 2H$.



Answer: (50 μF)

Solution:

Soln:- $\omega = 100 \text{ rad/s.}$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$100 = \sqrt{100^2 + (X_L - X_C)^2}$$

$$100^2 = 100^2 + (X_L - X_C)^2$$

$X_L = X_C$

$100 \times 2 = \frac{1}{100 \times C}$
 $C = \frac{1}{10^4 \times 2}$
 $\Rightarrow 0.5 \times 10^{-4}$
 $\Rightarrow \underline{50 \mu F}$

Question: Two blocks of mass 15 kg and 25 kg, moving towards each other with speeds 6 m/s and 10 m/s, undergo inelastic collision. If specific heat of both the blocks material is S then rise in temperature of blocks is equal to (Initially both blocks at same temperature).

Options:

- (a) 30/5
- (b) 15/5
- (c) 10/5
- (d) 20/5

Answer: (a)

Solution:

Before After

$\square \square \quad \square \square$

$$KE_{\text{loss}} = \frac{1}{2} \times m_L \times (u_{\text{rel}})^2 \times (1 - e^2)$$

$$= \frac{1}{2} \times \left[\frac{15 \times 25}{40} \right] \times (10)^2 \times 1$$

$$= 1200 \text{ J}$$

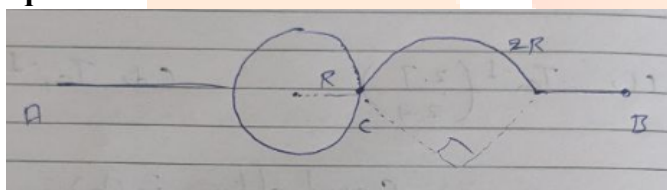
$$Q = m \Delta T$$

$$\therefore \Delta T = \frac{Q}{m} = \frac{1200}{40 \times 5} = \frac{30}{5}$$

$Q = m \Delta T$

$1200 = 40 \times 5 \Delta T$

Question: A wire with uniform resistance for unit length λ , bent into a circle of radius R is attached with a quadrant whose length is $2R$ as shown in figure. Find the equivalent resistance between A & B.

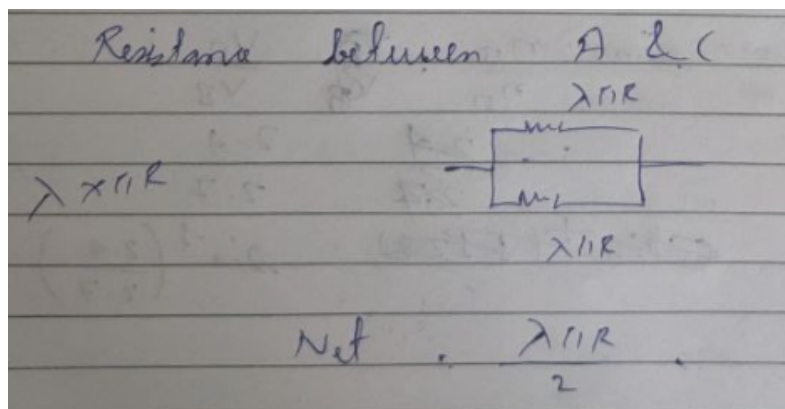


Options:

- (a) $2\lambda R \left(1 + \frac{\pi}{4} \right)$
- (b) λR
- (c) $\lambda R \left(1 + \frac{\pi}{4} \right)$
- (d) $\frac{\lambda R}{4}$

Answer: (a)

Solution:



$\frac{\lambda R}{2}$

$\frac{\lambda R}{2} + 2R$

$2R \left(1 + \frac{R}{A} \right)$

Question: A simple pendulum of length 3 m completes 20 oscillations in 10 s. Another simple pendulum completes 40 oscillations in the same time. Find the length of the second pendulum.

Options:

- (a) 0.75 m
- (b) 1.5 m
- (c) 6 m
- (d) 12 m

Answer: (a)

Solution:

$$n = \frac{t}{T} \Rightarrow n \propto \frac{1}{T}$$

For a simple pendulum:

$$T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow T \propto \sqrt{L}$$

Hence,

$$n \propto \frac{1}{\sqrt{L}}$$

So,

$$\frac{n_1}{n_2} = \sqrt{\frac{L_2}{L_1}}$$

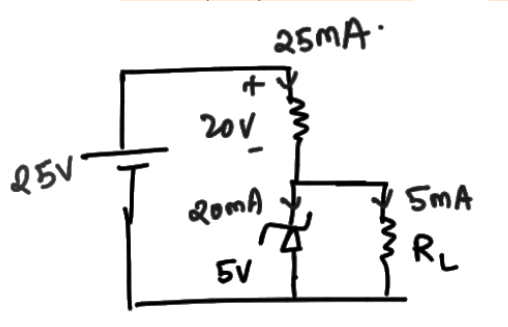
Substitute values:

$$\frac{20}{40} = \sqrt{\frac{L_2}{3}} \Rightarrow \frac{1}{2} = \sqrt{\frac{L_2}{3}}$$

Squaring:

$$\frac{1}{4} = \frac{L_2}{3} \Rightarrow L_2 = \frac{3}{4} = 0.75 \text{ m}$$

Question: Max. Current through the load can be 5 mA Whereas Current through Zener diode can be max 4 times of load current find the Value of Series resistance to be Connected. ($V_{\text{max(Zener)}} = 5\text{V}$).



Answer: (800Ω)

Solution:

$$R_L = \frac{5}{5} \times 10^3 \Rightarrow 1000\Omega = 1\text{k}\Omega$$

$$R_s = \frac{20}{25} \times 10^3 \Rightarrow 0.8 \times 10^3 \Rightarrow 800\Omega$$

Question: An electromagnetic wave is propagating in free space. The electric field associated with the wave is given by $E(t) = 300\sin(2 \times 10^8 t) \text{ Vm}^{-1}$. Find the average power per unit area carried by the wave.

Answer: (120)**Solution:**

Write the answer correct to the nearest integer.

$$(\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}, c = 3 \times 10^8 \text{ m s}^{-1})$$

Average intensity (power per unit area) of an EM wave:

$$I_{\text{avg}} = \frac{1}{2} \epsilon_0 c E_0^2$$

Here,

$$E_0 = 300 \text{ V m}^{-1}$$

$$I_{\text{avg}} = \frac{1}{2} \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times (300)^2$$

$$I_{\text{avg}} = 119.5 \text{ W m}^{-2}$$

Final Numerical Answer

120

Question: Assertion (A): Ferromagnetic substances are divided into a large number of small regions called magnetic domains, each of which is spontaneously magnetised.**Reason (R):** Above the Curie temperature, thermal agitation overcomes the tendency of magnetic moments to align, causing the disappearance of ferromagnetism.**Options:**

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

Answer: (b)**Question: Two Spring mass systems suspend masses of 100 gm and 200 gm respectively, in equilibrium. Energy of first block is E, Find the Energy of second Block. ($4K_a = 3K_b$).****Options:**

- (a) $16/3 E$
- (b) $4/5 E$
- (c) $5/7 E$
- (d) $11/7 E$

Answer: (a)**Solution:**

Sol:-

$$K_A x_A = m_A g$$

$$K_A x_A = 100 \times 10^{-3} \times g$$

$$x_A = \frac{10^{-1} \times g}{K_A}$$

$$E_A = \frac{1}{2} K_A \times \left(\frac{10^{-1} \times g}{K_A} \right)^2$$

$$\Rightarrow \frac{1}{2} \times \left(\frac{10^{-2} \times g^2}{K_A} \right) = e$$

U

$$K_B x_B = 200 \times 10^{-3} \times g$$

$$x_B = \frac{2 \times 10^{-1} \times g}{K_B}$$

$$E_B = \frac{1}{2} \times K_B \times \left(\frac{2 \times 10^{-1} \times g}{K_B} \right)^2$$

$$\Rightarrow \frac{1}{2} \times \frac{4 \times 10^{-2} \times g^2 \times 4}{3 K_A}$$

$$\Rightarrow \frac{1}{2} \left(\frac{10^{-2} \times g^2}{K_A} \right) \times \frac{4 \times 4}{3}$$

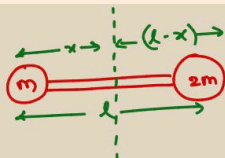
$$\Rightarrow \frac{16}{3} e$$

Question: Two masses m and $2m$ are attached at the two ends of a massless rod. If the angular momentum of the system is L about CoM, then find the value of ω .

Answer: ($\omega = \frac{3L}{2ml^2}$)

Solution:

Solution:-



Position of the com will be:

$$mx = 2m(l-x)$$

$$mx = 2ml - 2mx$$

$$3mx = 2ml$$

$$x = \left(\frac{2l}{3} \right) \rightarrow \text{from mass (m)}$$

and from mass ($2m$) = $l - \frac{2l}{3} = \left(\frac{l}{3} \right)$

MOI of the entire system about the com;

$$I_1 = m \times \left(\frac{2l}{3} \right)^2$$

$$\Rightarrow m \times \frac{4l^2}{9} \Rightarrow \frac{4ml^2}{9}$$

$$I_2 = 2m \times \left(\frac{l}{3} \right)^2$$

$$\Rightarrow 2m \times \frac{l^2}{9} \Rightarrow \frac{2ml^2}{9}$$

$$L = \left(\frac{4ml^2}{9} + \frac{2ml^2}{9} \right) \times \omega$$

$$L = \frac{2ml^2}{3} \omega \Rightarrow \omega = \frac{3L}{2ml^2}$$

Rotation about com

JEE-Main-23-01-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

Question: The correct order of ionisation energy of Cl, S, P, Al, Si is

Options:

- (a) $\text{Cl} > \text{P} > \text{S} > \text{Si} > \text{Al}$
- (b) $\text{P} > \text{Cl} > \text{S} > \text{Al} > \text{Si}$
- (c) $\text{Cl} > \text{S} > \text{P} > \text{Si} > \text{Al}$
- (d) $\text{Cl} > \text{Al} > \text{Si} > \text{P} > \text{S}$

Answer: (a)

Question: Given below are two statements

S-I: $[\text{CoBr}_4]^{2-}$ absorbs lesser energy than $[\text{CoCl}_4]^{2-}$

S-II: $[\text{CoCl}_4]^{2-}$ has higher crystal field splitting energy than $[\text{CoBr}_4]^{2-}$

Options:

- (a) Both S-I and S-II are correct
- (b) Both S-I and S-II are incorrect
- (c) S-I is correct and S-II are incorrect
- (d) S-I is incorrect and S-II are correct

Answer: (a)

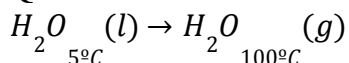
Question: Which of the following undergo nitration at fastest rate?

Options:

- (a) $\text{C}_6\text{H}_5\text{NO}_2$
- (b) $\text{C}_6\text{H}_5\text{CH}_3$
- (c) $\text{C}_6\text{H}_5\text{COOH}$
- (d) $\text{C}_6\text{H}_5\text{Br}$

Answer: (b)

Question: For the following change,



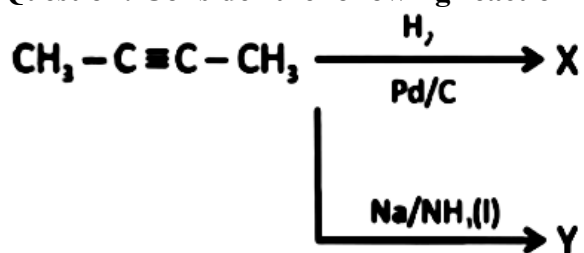
Select the correct answer:

Options:

- (a) $q = +ve, w = +ve, \Delta H = +ve$
- (b) $q = -ve, w = -ve, \Delta H = +ve$
- (c) $q = +ve, w = -ve, \Delta H = +ve$
- (d) $q = -ve, w = -ve, \Delta H = -ve$

Answer: (c)

Question: Consider the following reaction



Choose the correct option.

Options:

- (a) $\text{X} \Rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
 $\text{Y} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (cis)
- (b) $\text{X} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (cis)
 $\text{Y} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (trans)
- (c) $\text{X} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (cis)
 $\text{Y} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (cis)
- (d) $\text{X} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (trans)
 $\text{Y} \Rightarrow \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ (cis)

Answer: (b)

Question: Given below are two statements.

Statement-I: Sublimation is a purification technique that is used to separate those solid substances which changes from solid to vapour state without passing through liquid state.

Statement-II: If external atmospheric pressure is reduced, then boiling point of substance is decreased.

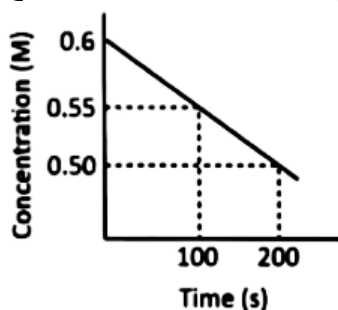
In the light of the above statements, choose the correct option.

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both Statement-I and Statement-II are incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (a)

Question: Consider the following graph of concentration vs time.



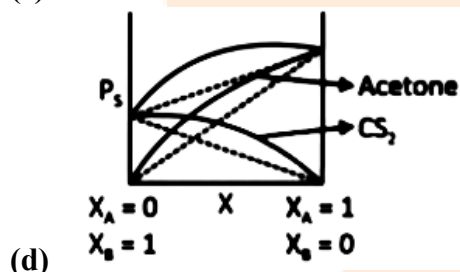
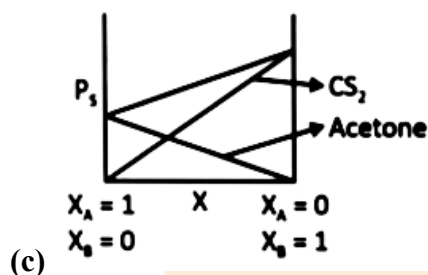
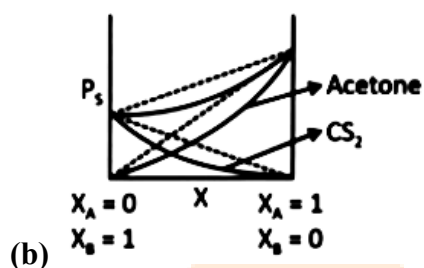
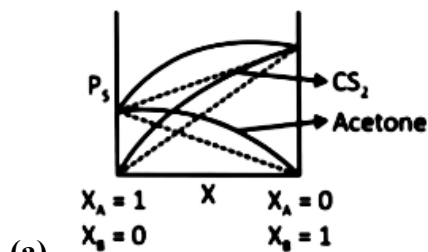
Find half-life of reaction.

Options:

- (a) 600 s
- (b) 200 s
- (c) 300 s
- (d) 100 s

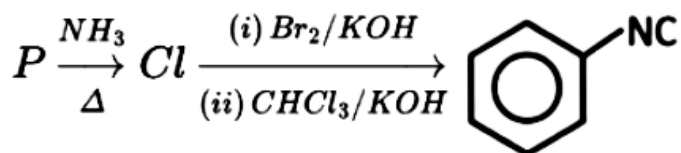
Answer: (a)

Question: A binary solution is formed by mixing Acetone (A) and CS₂ (B). The variation of vapour pressure v/s mole fraction will be
Options:



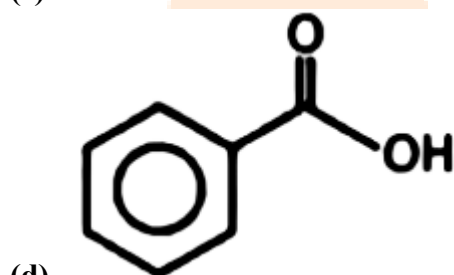
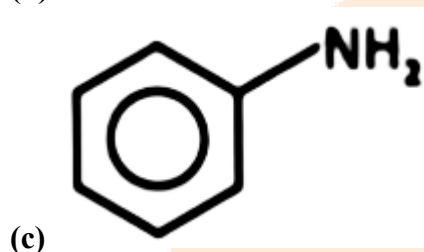
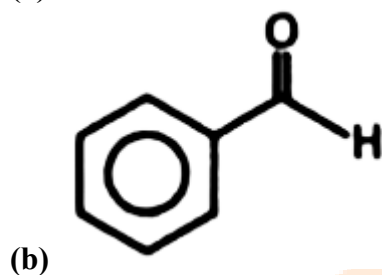
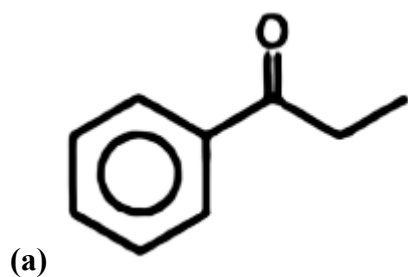
Answer: (a)

Question: Consider the following reaction:



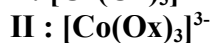
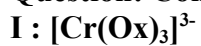
The structure of 'P' is

Options:



(d)
Answer: (d)

Question: Consider the two complexes



Find the ratio of CFSE of I to II complex (neglect pairing energy and consider Δ_0 for both complexes to be x)

Options:

(a) 2

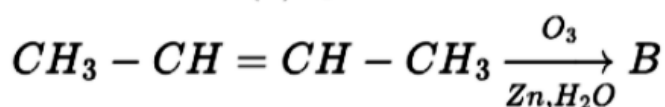
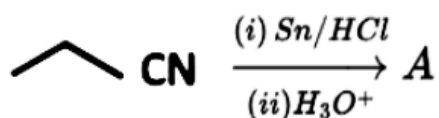
(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{2}{3}$

Answer: (b)

Question: Consider the following reactions



A and B are mixed and treated with dil. Base to give mixture of products.
Choose the incorrect product.

Options:

- (a)
- (b)
- (c)
- (d)

Answer: (c)

Question: Correct statement about 13th group.

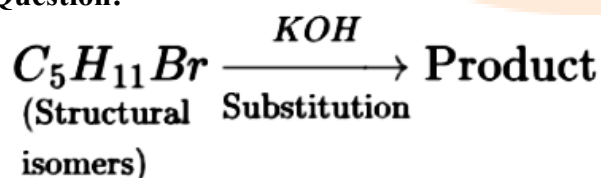
- A. Electronegativity decreases regularly down the group.
B. Ionic radii decrease down the group.
C. Boron has the highest ionisation energy.
D. Trichloride of aluminium are covalent in nature

Options:

- (a) A, B, C only
(b) C, D only
(c) A, C, D only
(d) B, C, D only

Answer: (c)

Question:



Given number of optically active isomer in product.

Options:

- (a) 6
(b) 2
(c) 5
(d) 0

Answer: (a)

Question: Match List-I with List-II.

List-I	List-II
A) Bayer's unsaturation test	i) Violet/purple colour
B) Ceric ammonium nitrate test of alcohols	ii) Red colour
C) Tollen's reagent test	iii) Silver mirror obtained
D) FeCl_3 test of phenol	iv) Pink colour discharge

Options:

- (a) A(ii), B(i), C(iv), D(iii)
 (b) A(ii), B(i), C(iii), D(iv)
 (c) A(iv), B(ii), C(iii), D(i)
 (d) A(iv), B(iii), C(ii), D(i)

Answer: (c)

Question: Consider the given cell



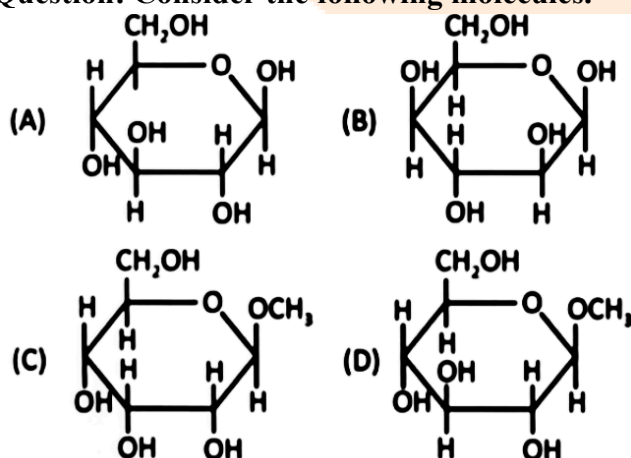
In which of the following cases, E_{cell} will increase

Options:

- (a) $[\text{Fe}^{2+}]$ increases
 $[\text{Cl}^-]$ increases
 (b) $[\text{Fe}^{2+}]$ increases
 $[\text{Cl}^-]$ decreases
 (c) $[\text{Fe}^{3+}]$ & $[\text{Cl}^-]$ increases
 (d) $[\text{Fe}^{2+}]$ decreases
 $[\text{Fe}^{3+}]$ decreases

Answer: (c)

Question: Consider the following molecules.



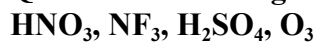
The examples of non-reducing sugar(s) are

Options:

- (a) A, B only
 (b) A, C only
 (c) B, D only
 (d) C, D only

Answer: (d)

Question: Following molecules are given:



Consider the molecule (T) having maximum number of lone pairs (on all atoms).

The bond angle ($\angle \text{XMX}$), where M is central atom in T is

Options:

- (a) 110°
 (b) 97°
 (c) 102°
 (d) 115°

Answer: (c)

Question: Consider the following statements and choose the correct option(s)

- A. Ni^{2+} forms pink colour complex with dimethylglyoxime ligand (dmg)
 B. Ni^{2+} complex with dmg contains two ring of five membered
 C. Ni^{2+} has 2 unpaired electron in d-orbitals in the complex
 D. Ni^{2+} complex with dmg is soluble at $\text{pH} = 9$

Options:

- (a) Only A & B
 (b) Only B & C
 (c) Only A & C
 (d) Only A, C and D

Answer: (a)

Question: Match the List-I with List-II and choose the correct option:

List-I	List-II
A) 2 nd orbit of He^+ ion	i) $-1.96 \times 10^{-17} \text{ J}$
B) 3 rd orbit of H-atom	ii) $-2.42 \times 10^{-19} \text{ J}$
C) 1 st orbit of Li^{2+} ion	iii) $-2.178 \times 10^{-18} \text{ J}$
D) 2 nd orbit of Li^{2+} ion	iv) $-4.9 \times 10^{-18} \text{ J}$

Options:

- (a) A-(iii), B-(ii), C-(i), D-(iv)
 (b) A-(iii), B-(ii), C-(iv), D-(i)
 (c) A-(iv), B-(iii), C-(ii), D-(i)
 (d) A-(i), B-(ii), C-(iii), D-(iv)

Answer: (a)

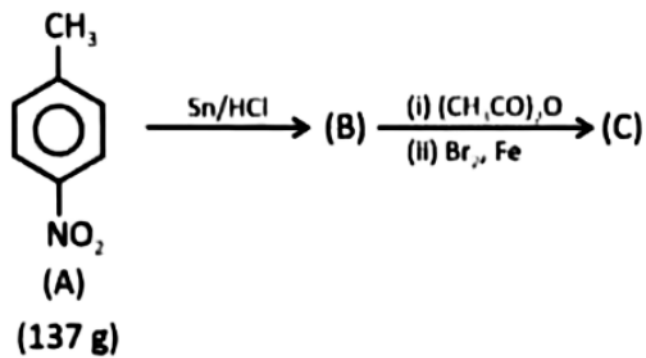
Question: x g of pure Cl_2 is reacted with $\text{Ba}(\text{OH})_2$ to form $\text{Ba}(\text{ClO}_3)_2$. $\text{Ba}(\text{OH})_2$ concentration is 1 M and volume is 25 mL. Find x.

Options:

- (a) 2
- (b) 6
- (c) 4
- (d) 1

Answer: (a)

Question: In the reaction sequence, what is the mass (in grams) of product (C) formed?



Options:

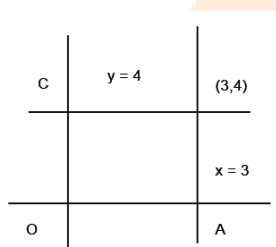
Answer: (228)

JEE-Main-23-01-2026 (Memory Based)

[MORNING SHIFT]

Maths

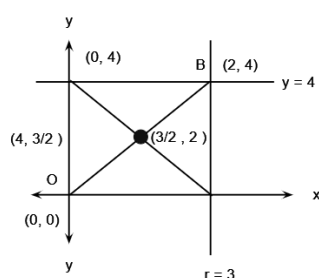
Question: A rectangle is formed by lines $x = 0$, $y = 0$, $x = 3$, $y = 4$. A line perpendicular to $3x + 4y + 6 = 0$ divides the rectangle into two equal parts, then the distance of the line from $\left(-1, \frac{3}{2}\right)$ is



Options:

- (a) 2
- (b) $17/10$
- (c) $6/5$
- (d) $8/5$

Answer: (b)



$$3x + 4y + 6 = 0, \quad m_1 = -\frac{3}{4}$$

$$\text{eq}^n \text{ of } AC, \quad m_2 = \frac{4}{3}$$

$$y - 2 = \frac{4}{3} \left(x - \frac{3}{2} \right)$$

$$3y - 6 = 4x - 6$$

$$4x - 3y = 0$$

$$D = \left| \frac{4(-1) - 3\left(\frac{3}{2}\right)}{\sqrt{16+9}} \right| = \left| \frac{-4 - \frac{9}{2}}{5} \right| = \frac{17}{10}$$

Question: Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$ and R be a relation R , such that $R = \{(x, y): (2x + y) \leq -2, x \in A, y \in A\}$. Let f = number of elements in R m = minimum number of elements to be added in R to make reflexive. N = minimum number of elements to be added to make it symmetric, then $(f + m + n)$ is

Options:

- (a) 10
- (b) 17
- (c) 11
- (d) 14

Answer: (b)

Given

$$A = \{-2, -1, 0, 1, 2, 3, 4\}$$

Relation

$$R = \{(x, y) \mid 2x + y \leq -2, x \in A, y \in A\}$$

$$\begin{aligned} \text{Number of elements in } R &= (-2, -2) (-2, -1) (-2, 0) (-2, 1) (-2, 2) \\ &\quad (-1, 0) (0, -2) (-1, -1) (-1, -2) \\ l &= 9 \end{aligned}$$

2. Elements to add to make R reflexive (call it m)

Reflexive $\Rightarrow$ all (a, a) must be in R .

Check $2a + a = 3a \leq -2$:

- $(-2, -2)$ ✓
- $(-1, -1)$ ✓
- $(0, 0), (1, 1), (2, 2), (3, 3), (4, 4)$ ✗

Already present: 2

Total needed: 7

$$m = 7 - 2 = 5$$

3. Elements to add to make R symmetric (call it n)

For symmetry, if $(x, y) \in R$, then (y, x) must also be in R .

Missing reverse pairs are:

- $(1, -2)$ for $(-2, 1)$
- $(2, -2)$ for $(-2, 2)$
- $(0, -1)$ for $(-1, 0)$

$$n = 3$$

$$l + m + n = 9 + 5 + 3 = 17$$

Question: $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{1}{1 + (\tan 2x)^{1/3}}$
Options:

- (a) $\frac{\pi}{24}$
- (b) $\frac{\pi}{12}$
- (c) $\frac{\pi}{6}$
- (d) $\frac{\pi}{48}$

Answer: (b)

$$I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan 2x}} \dots\dots(1)$$

$$I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan 2\left(\frac{\pi}{24} + \frac{5\pi}{24} - x\right)}}$$

$$I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan \left(\frac{\pi}{2} - 2x\right)}}$$

$$I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\cot 2x}}$$

$$I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{\sqrt[3]{\tan 2x}}{1 + \sqrt[3]{\tan 2x}} dx \dots\dots(2)$$

$$(1) + (2)$$

$$\Rightarrow 2I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} 1 dx$$

$$2I = [x]_{\frac{\pi}{24}}^{\frac{5\pi}{24}}$$

$$2I = \frac{5\pi}{24} - \frac{\pi}{24} = \frac{\pi}{6}$$

$$I = \frac{\pi}{12}$$

Question: Number of 4 letters words with or without meaning formed from the letters of the word PQRSSSTTUVV is

Options:

- (a) 1232
- (b) 1400
- (c) 1422
- (d) 1162

Answer: (c)

PQRSSSTTUVV

PQR 3S 2T UV.V

All alike

$${}^7C_4 \times 4!$$

$$= 840$$

Two alike two lllle

$${}^6C_2 \times {}^3C_1 \times \frac{4!}{2} = 1S \times 3 \times 12 = 540$$

$$1. {}^6C_1 \times \frac{4!}{3} = 24$$

$${}^3C_2 \frac{4!}{2! \times 8!} = 3 \times 6 = 18$$

$$840 + 540 + 18 + 24 = 1422$$

Question: Find the value of

$$\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \dots + \frac{{}^{100}C_{100}}{101}$$

Options:

- (a) $\frac{2^{100}}{100}$
- (b) $\frac{101}{2^{100}}$
- (c) $\frac{101}{2^{101}}$
- (d) $\frac{100}{2^{101}}$

Answer: (c)

$$\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \dots + \frac{{}^{100}C_{100}}{101}$$

$$\left[\frac{{}^nC_r}{n} = \frac{{}^{n-1}C_{r-1}}{r} \right] n - 1 = 100$$

$$n = 101$$

$$\frac{1}{101} \left({}^{101}C_{31} + {}^{101}C_{52} + \dots + {}^{101}C_{101} \right)$$

$$\frac{1}{101} \left(\frac{2^{101}}{2} \right)$$

$$= \frac{2^{100}}{101}$$

Question: $S = \{1, 2, \dots, 100\}$ Select two different no. m, n such that $n - m \geq 10$ is having Probability $\frac{a}{b}$ Find $a + b$ where a and b are Coprime.

Options:

- (a) 200
- (b) 201
- (c) 202
- (d) 203

Answer: (b)

$$S = \{1, 2, \dots, 100\}$$

Select two number m, n s.t. $n - m \geq 10$

have probability $= \frac{a}{b}$

$\therefore$ if $m = 1$, for $n \rightarrow 90$ choices

if $m = 2$, for $n \rightarrow 89$ choices

if $m = 3$, for $n \rightarrow 88$ choices

. . . .

if $m = 90$, for $n \rightarrow 1$ choice

Total ways to select m, n, s that

$n - m \geq 10$ are $90 + 89 + \dots + 1$

$$= \frac{90+91}{2}$$

Total number of ways to select two numbers

$$m, n \text{ are } \frac{100 \times 99}{2} = {}^{100}C_2$$

$$\therefore \text{Probability} = \frac{\frac{90+91}{2}}{\frac{100 \times 99}{2}} = \frac{91}{10 \times 11} = \frac{91}{110}$$

$$\therefore a = 91, b = 110$$

$$\Rightarrow a + b = 110 + 91 = 201$$

Question: $x^4 dy + (4x^3 y + 2 \sin x) dx = 0$ & $f\left(\frac{\pi}{2}\right) = 0$ then $\pi^4 f\left(\frac{\pi}{3}\right) = ?$

Options:

- (a) 3
- (b) 3^2
- (c) 3^3
- (d) 3^4

Answer: (d)

$$x^4 dy + (4x^3 y + 2 \sin x) dx = 0 \quad f\left(\frac{\pi}{4}\right) = 0$$

$$\tan \pi^4 f\left(\frac{\pi}{3}\right) =$$

$$\frac{x^4}{x^4} \frac{dy}{dx} + \frac{4x^3 y}{x^4} = - \frac{\sin x}{x^4}$$

$$\frac{dy}{dx} + \frac{y}{x} = \frac{-2 \sin x}{x^4}$$

$$I.f = e^{\ln x^4} = x^4$$

$$yx^4 = [2 \cos x] + c$$

$$c = 0$$

$$f\left(\frac{\pi}{3}\right) + f(x) = \frac{2 \cos x}{x^4}$$

$$\pi^4 f\left(\frac{\pi}{3}\right) = \frac{1}{\pi^4} 3^4 \times \pi^4 = 81$$

Question: If A is a matrix of order 3 and $x = |3 \operatorname{adj}(A^2) \cdot \operatorname{adj}(2A)|$ and $|A| = 6$ and $x = 2^n \cdot 3^m$, then $m + n$ is

Options:

- (a) 21
- (b) 25
- (c) 27
- (d) 19

Answer: (a)

$$x = |3 \operatorname{adj}(A^2) \cdot \operatorname{adj}(2A)|$$

For 3×3 matrix:

$$\operatorname{adj}(A^2) = \operatorname{adj}(A)^2, \quad \operatorname{adj}(2A) = 2^2 \operatorname{adj}(A) = 4 \operatorname{adj}(A)$$

$$x = |12 \operatorname{adj}(A)^3| = 12^3 |\operatorname{adj}(A)|^3$$

$$|\operatorname{adj}(A)| = |A|^2 = 6^2 = 36$$

$$x = (12 \cdot 36)^3 = 432^3$$

$$432 = 2^4 \cdot 3^3 \Rightarrow x = 2^{12} \cdot 3^9$$

$$n = 12, m = 9 \Rightarrow \boxed{m + n = 21}$$

Question: For given vectors $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ where

$\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{b}$. Then the value of $\left(\frac{\vec{a} - \vec{b}}{|\vec{a} - \vec{b}|}\right) \cdot \vec{d}$ is

Options:

- (a) -35
- (b) 53
- (c) -52
- (d) 25

Answer: (a)

$$\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$$

$$\vec{c} = \vec{a} \times \vec{b}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix}$$

$$= 3\hat{i} + 5\hat{j} - \hat{k}$$

$$\vec{d} = \vec{c} \times \vec{b}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & -1 \\ 2 & -1 & 1 \end{vmatrix}$$

$$= 4\hat{i} - 5\hat{j} - 13\hat{k}$$

$$(\vec{a} - \vec{b}) \cdot \vec{d}$$

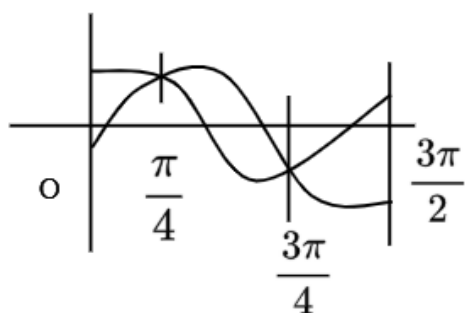
$$= (-3\hat{i} + 2\hat{j} + \hat{k}) (4\hat{i} - 5\hat{j} - 13\hat{k})$$

$$= -12 - 10 - 13$$

$$= -35$$

Question: $f(x) = \max(\sin x, \cos x)$. Find area bounded by $f(x)$ and x -axis and $x = 0$ and $x = \frac{3\pi}{2}$.

Answer: $(2\sqrt{2} - 1)$



$$\int_0^{\frac{\pi}{4}} \cos x + \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sin x + \int_{\frac{3\pi}{4}}^{\frac{3\pi}{2}} \cos x$$

$$[\sin x]_0^{\frac{3\pi}{4}} [-\cos x]_{\frac{\pi}{4}}^{\frac{5\pi}{4}} + [\sin x]_{\frac{3\pi}{4}}^{\frac{3\pi}{2}}$$

$$\frac{1}{\sqrt{2}} + - \left[-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right] + \left[-1 - \frac{1}{\sqrt{2}} \right]$$

$$\left(\frac{1}{\sqrt{2}} + \sqrt{2} - 1 + \frac{1}{\sqrt{2}} \right) = 2\sqrt{2} - 1$$

Question: The line $y = x+1$ intersects the ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$ at A and B. Find the angle subtended by segment AB and centre of ellipse is

Options:

(a) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$

(b) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$

(c) $\frac{\pi}{2} + 2\tan^{-1}\left(\frac{1}{4}\right)$

(d) $\frac{\pi}{4} + \tan^{-1}\left(\frac{1}{4}\right)$

Answer: (a)

$$y = n + 1, \quad \frac{n^2}{2} + \frac{y^2}{1} = 1$$

$$y = n + 1$$

$$\frac{x^2}{2} + \frac{(x+1)^2}{1} = 1$$

$$x^2 + 2(n^2 + 1 + 24) = 2$$

$$3x^2 + 4x = 0$$

$$x = 0, \quad -\frac{4}{3}$$

$$\Rightarrow y = -\frac{4}{3} + 1$$

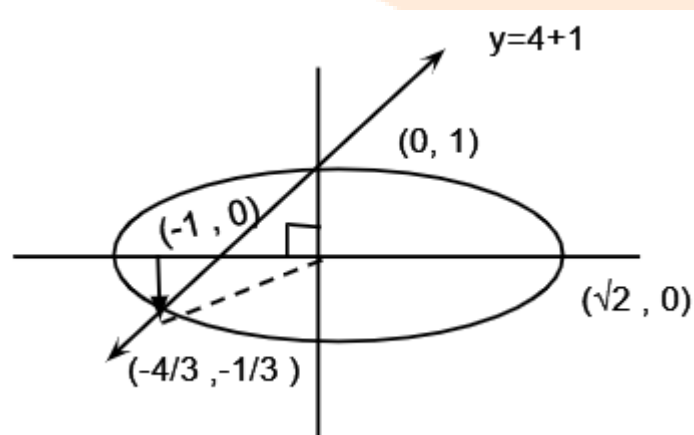
$$y = -\frac{1}{3}$$

$$\tan \theta = \frac{-\frac{1}{3}}{-\frac{4}{3}} = \frac{1}{4}$$

$$\tan \theta = \frac{1}{4}$$

$$\theta = \tan^{-1}\left(\frac{1}{4}\right)$$

$$\Rightarrow \frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$$



$$\text{If } \begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$$

Question:

then value of $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$ is -

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (a)

$$\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$$

$$1(1 - \cos^2 \gamma) - \cos \alpha (\cos \alpha - \cos \gamma \cos \beta) + \cos \beta (\cos \alpha \cos \gamma - \cos \beta)$$

$$= 0 - \cos \alpha (0 - \cos \gamma \cos \beta) + \cos \beta (\cos \alpha \cos \gamma - 0)$$

$$1 - \cos^2 \gamma - \cos^2 \alpha - \cos^2 \beta = 0$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Question: Let the domain of $f(x) = \log_3(\log_5(\log_7(9x - x^2 - 13)))$ is (m, n) . Let

$\frac{n}{3}$ and $\frac{3m}{8}$ be eccentricity and length of latus rectum of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

respectively, then the value of $\left(\frac{a+b^2}{a-b^2}\right)$ is equal to

Options:

- (a) 7
- (b) 9
- (c) 3
- (d) 13

Answer: (a)

$$9x - x^2 - 13 > 7$$

$$\Rightarrow x^2 - 9x + 20 < 0$$

$$(x - 5)(x - 4) < 0 \quad x \in (4, 5)$$

$$m = 4, \quad n = 5$$

$$e = \frac{5}{3}, \quad \frac{2b^2}{a} = \frac{3}{8} \times 4 = \frac{3}{2}$$

$$\frac{b^2}{a} = \frac{3}{4}$$

$$\frac{a}{b^2} = \frac{4}{3} \Rightarrow \frac{a+b^2}{a-b^2} = \frac{4+3}{4-3} = 7$$

Question: $(f(x))^2 = 25 + \int_0^x ((f(t))^2 + (f'(t))^2) dt, f(x) > 0,$

Find average value of $f(\ln 1), f(\ln 1), \dots, f(\ln 625)$

Answer: ()

Question: The number of solutions of $13 \cos 2\theta + 8 \cos \theta - 3\sqrt{3} = 0$ if $\theta \in [-3\pi, 2\pi]$ is

Options:

- (a) 4
- (b) 3
- (c) 0
- (d) 5

Answer: (d)

$$13 \cos 2\theta + 8 \cos \theta - 3\sqrt{3} = 0,$$

$$\Rightarrow 13(2\cos^2\theta - 1) + 8 \cos \theta - 3\sqrt{3} = 0$$

$$26 \cos^2 \theta + 8 \cos \theta - (13 + 3\sqrt{3}) = 0$$

$$\text{let } \cos \theta = x$$

$$\Rightarrow 26x^2 + 8x - (13 + 3\sqrt{3}) = 0$$

$$x = \frac{-8 \pm \sqrt{8^2 + 4 \times 26(13 + 3\sqrt{3})}}{52}$$

$$x = \frac{-8 \pm \sqrt{64 + 104(13 + 3\sqrt{3})}}{52}$$

$$x = \frac{-8 \pm 44.22}{52}$$

$\therefore$ Only one +ve root possible as $-1 \leq \cos \theta \leq 1$

$\therefore$ On $[-3\pi, 2\pi]$, 5 solution possible

Question: B(4, 10, α) and C are points on line $\frac{x}{1} = \frac{y-2}{2} = \frac{z+6}{2}$ such that BC = 10. If point A is (2, 3, 4) area of triangle ABC is

Answer: ($5\sqrt{41}$)

Line:

$$\frac{x}{1} = \frac{y-2}{2} = \frac{z+6}{2}$$

Point B(4, 10, α) lies on the line.

Let the common ratio be t .

$$x = t \Rightarrow t = 4$$

$$z = 2t - 6 = 2$$

$$\boxed{\alpha = 2} \Rightarrow B = (4, 10, 2)$$

Direction vector of line:

$$\vec{d} = (1, 2, 2)$$

$$A = (2, 3, 4) \Rightarrow \vec{BA} = (-2, -7, 2)$$

Distance of point A from line:

$$\text{Distance} = \frac{|\vec{BA} \times \vec{d}|}{|\vec{d}|} = \frac{3\sqrt{41}}{3} = \sqrt{41}$$

Area of $\triangle ABC$

$$\text{Area} = \frac{1}{2} \times 10 \times \sqrt{41}$$

$$\boxed{\text{Area} = 5\sqrt{41}}$$

$$\int \frac{(2-x^2)e^x}{\sqrt{1+x} \cdot (\sqrt{1-x})^3} dx =$$

Question:

Answer: $\left(\frac{1}{(1-x)\sqrt{1-x^2}} \right)$

$$\int \frac{e^x(2-x^2)}{\sqrt{1+x} \cdot (\sqrt{1-x})^3} dx$$

$$= \int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx$$

$$= \int e^x \left(\frac{1+(1-x^2)}{(1-x)\sqrt{1-x^2}} \right) dx$$

$$= \int e^x \left(\frac{1}{(1-x)(\sqrt{1-x^2})} + \frac{1+x}{\sqrt{1-x^2}} \right) dx$$

$$= \int e^x \left(\frac{1}{(1-x)\sqrt{1-x^2}} + \frac{\sqrt{1+x}}{\sqrt{1-x}} \right) dx$$

$$e^x \cdot \sqrt{\frac{1+x}{1-x}} + c$$

$$f(x) = \frac{\sqrt{1+x}}{\sqrt{1-x}}$$

$$f'(x) = \frac{\sqrt{1-x} \cdot \frac{1}{2\sqrt{1+x}} + \sqrt{1+x} \cdot \frac{1}{2\sqrt{1-x}}}{(1-x)}$$

$$= \frac{1-x+(1+x)}{2(1-x)\sqrt{1-x^2}} = \frac{1}{(1-x)\sqrt{1-x^2}}$$

Question: If mean and variance of -10, -7, -1, x, y, 9, 6, 2 are 2 and 51 respectively then mean of x, y, (x-y), $|10(x-2y)|$ is (where $x > y$)

Options:

- (a) 10
(b) 4
(c) 44
(d) 11

Answer: (d)

Given data - $-10, -7, -1, x, y, 9, 6, 2$

mean $(\bar{x}) = 2$, Variance $(\sigma^2) = 51$

$$\bar{x} = \frac{-10-7-1+x+y+9+6+2}{8}$$

$$2 = \frac{-18+x+y+17}{8}$$

$$x + y = 17 \dots\dots(1)$$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$51 = \frac{(-12)^2 + (-9)^2 + (-3)^2 + (x-2)^2 + (y-2)^2 + 7^2 + 4^2 + 0^2}{8}$$

$$\Rightarrow (x-2)^2 + (y-2)^2 = 109$$

$$\text{by (1)} (x-2)^2 + (15-x)^2 = 109$$

$$2x^2 = 34x + 120 = 0$$

$$\Rightarrow x^2 - 17x + 60 = 0$$

$$\Rightarrow x = 5 \text{ or } 12$$

$$x = 12, y = 5 \text{ as } x > y$$

$$\therefore \text{Mean of } x, y, (x-y), 10(x-2y)$$

$$i.e., 12, 5, 7, 20$$

$$\text{Required mean} = \frac{12+5+7+20}{4}$$

$$= 11$$

$$f(x) = \begin{cases} \frac{ax^2+2ax+3}{4x^2+4x-3} & x \neq \frac{3}{2}, \frac{1}{2} \\ b & x = -\frac{3}{2}, \frac{1}{2} \end{cases}$$

Question:

If $f(x)$ is continuous at

$$x = -\frac{3}{2}, \text{ then find } x \text{ if } f(f(x)) = \frac{7}{5}.$$

Options:

- (a) 1
(b) -1
(c) 4
(d) -4

Answer: (a)

Denominator:

$$4x^2 + 4x - 3 = (2x - 1)(2x + 3)$$

For continuity, numerator must vanish at $x = -\frac{3}{2}$:

$$a\left(\frac{9}{4}\right) + 2a\left(-\frac{3}{2}\right) + 3 = 0$$

$$\frac{9a}{4} - 3a + 3 = 0 \Rightarrow -\frac{3a}{4} + 3 = 0 \Rightarrow a = 4$$

Then

$$f(x) = \frac{4x^2 + 8x + 3}{(2x - 1)(2x + 3)} = \frac{(2x + 1)(2x + 3)}{(2x - 1)(2x + 3)} = \frac{2x + 1}{2x - 1}$$

$$f(x) = \frac{2x + 1}{2x - 1}$$

$$f(f(x)) = \frac{2\left(\frac{2x+1}{2x-1}\right) + 1}{2\left(\frac{2x+1}{2x-1}\right) - 1} = \frac{6x + 1}{2x + 3}$$

$$\frac{6x + 1}{2x + 3} = \frac{7}{5}$$

$$5(6x + 1) = 7(2x + 3) \Rightarrow 30x + 5 = 14x + 21 \Rightarrow 16x = 16 \Rightarrow x = 1$$

Question: If coefficients of x , x^2 and x^3 are in arithmetic progression of the binomial expansion of $(1 + x^2)^2 (1+x)^n$, $n \in \mathbb{N}$. Then sum of all values of n is

Options:

- (a) 9
- (b) 7
- (c) 8
- (d) 10

Answer: (a)

$$(1 + x^2)^2(1 + x)^n = (1 + 2x^2 + x^4)(1 + x)^n$$

Coefficients:

- of x : $a_1 = n$
- of x^2 : $a_2 = \binom{n}{2} + 2$
- of x^3 : $a_3 = \binom{n}{3} + 2n$

AP condition:

$$2a_2 = a_1 + a_3$$

$$2\left(\binom{n}{2} + 2\right) = n + \binom{n}{3} + 2n$$

$$n(n-1) + 4 = 3n + \frac{n(n-1)(n-2)}{6}$$

$$n^3 - 9n^2 + 20n - 24 = 0$$

$$n = 2, 3, 4$$

Sum of all $n = 2 + 3 + 4 = 9$
